

Pipeline Optimisation — Measured Impact

PCSMCPL Change Detection · benchmarked 13 Aug 2026 · NVIDIA RTX 4050 Laptop (6 GB), Windows 11

1 · METHOD

Every figure below is a wall-clock measurement on the production working grid (6144 × 6144 px, 37.7 Mpx) taken by running the same functions twice on *identical inputs* — once with `GPU_IMAGE_OPS=true`, once with it off so the original `scipy / skimage` code paths run. Timings are CUDA-synchronised; the GPU pass is warmed once before measuring. Numbers marked (*scaled*) are extrapolated from a measured per-unit cost and are labelled as such.

2 · PERFORMANCE

2.1 Measured stage costs, per analysis

Stage	Before (s)	After (s)	Speed-up	Saved (s)
Histogram matching (co-registration)	12.040	1.145	10.5×	10.895
Instance stage, end-to-end	22.954	3.586	6.4×	19.368
Total changed portion	34.994	4.731	7.40×	30.263

The instance stage is reported end-to-end rather than as a sum of its parts, to avoid double-counting. Its internal components measured: structural change field 15.316 → 0.350 s (43.7×), colour change field 3.108 → 0.253 s (12.3×), binary closing 0.458 → 0.111 s (4.1×), binary dilation 0.277 → 0.066 s (4.2×).

2.2 Whole-run effect

Absolute saving is fixed at **30.3 s per analysis**. The *ratio* depends on the unchanged remainder — segmentation, warping, SAM refinement, vectorisation, I/O — which was already GPU-bound and is untouched by this work. By Amdahl's law:

$$S = (T_{\text{other}} + 34.994) / (T_{\text{other}} + 4.731)$$

T_{other} = wall-clock of all stages this work did not modify. Measured anchor: ChangeStar segmentation costs 0.451 s/tile × 64 tiles × 2 epochs = **57.7 s (scaled)**.

T _{other} assumption	Before (s)	After (s)	Speed-up	Run time cut
57.7 s — segmentation only (lower bound)	92.7	62.4	1.48×	32.6 %
80 s — + land cover	115.0	84.7	1.36×	26.3 %
110 s — + SAM refinement	145.0	114.7	1.26×	20.9 %
150 s — + vectorise & I/O	185.0	154.7	1.20×	16.4 %

So the defensible claim is a **1.2–1.5× reduction in total analysis wall-clock**, and **7.4× on the portion actually optimised**.

2.3 Secondary gains

- CD-mode backend** (ResNet-18 DCVA, non-default path) ran entirely on CPU. Now GPU + fp16: 0.256 → 0.063 s per 8-chip batch (4.1×); (*scaled*) to a 6144² grid's 1024 chips, 32.8 → 8.0 s.
- Host RAM**. Land-cover stacks retained as fp16: 2.11 → 1.06 GB. Histogram matching no longer argsorts both epochs: ≈ 0.9 GB of temporaries removed.
- Disk**. Ingest wrote a full-size intermediate, read it back, then wrote the COG. Rasters under ≈ 700 Mpx now stage in RAM — one write and one read eliminated. GDAL block cache raised from its 5 % default to 2 764 MB.

3 · ACCURACY

The performance work changed detection accuracy by exactly zero — by construction, and verified. On a full synthetic analysis run with the GPU path on and off: change-probability map **max |Δ| = 0.000**, identical polygon

count, identical areas and confidences, identical diagnostics. **No pixel anywhere crosses the 0.5 decision threshold differently.**

Component-level equivalence checks:

Operation	Agreement with original	Verification
Binary dilation / erosion / closing	Bit-identical	vs <code>scipy.ndimage</code> , random + structured masks, 1–3 iterations
Histogram matching	Byte-identical	vs <code>skimage</code> , dense and sparse histograms
Change fields (colour, structural)	$\max \Delta \approx 5 \times 10^{-6}$	float32 rounding; 0 threshold crossings

Border handling was the one place equivalence had to be earned: torch's `reflect` padding is scipy's `mirror`. The first implementation diverged by up to 0.77 after normalisation, with 100 % of differing pixels inside the frame halo. Replacing it with an explicit symmetric pad removed the difference entirely.

3.1 Where accuracy gains actually came from

Accuracy improved earlier in the project, from model and logic changes — not from this optimisation. Recorded measurements, by proxy:

Change	Metric	Before	After
Segmenter: U-Net → ChangeStar ViT-B	BEFORE/AFTER footprint IoU	0.112	0.554
Co-registration guard (fabricated shift removed)	Grid alignment IoU	0.114	1.000
Colour index → learned land cover	Scene falsely read as vegetation loss	8.5 %	eliminated

What is not claimed. There is no labelled ground-truth set for this area, so **precision, recall and F1 are unmeasured** and no percentage accuracy improvement can honestly be quoted. Footprint IoU is a measure of *input quality* — how far the two epochs' segmentations agree — and is a proxy, not a detection score. Converting these into a defensible accuracy figure requires an officer-verified benchmark; the review loop already exports exactly that data, and once ≈ 150 labelled polygons exist the learned decider can be scored against the rule cascade.

4 · RESOURCE ENVELOPE

Resource	Budget	Measured peak	Enforcement
VRAM	4.0 GB	1.59 GB	<code>torch</code> memory fraction + <code>ORT gpu_mem_limit</code> ; 5 GB alloc refused with 4.96 GB free
Host RAM	18.0 GB	—	sizes the working grid and GDAL cache (advisory, not an rlimit)

5 · LIMITATIONS

- Timings use synthetic imagery of production dimensions. Structure affects the instance stage (candidate count) but not the per-pixel work, which dominates.
- Single machine, single run per configuration; no variance reported.
- Segmentation total (57.7 s) and CD-backend total (32.8 / 8.0 s) are scaled from measured per-tile and per-batch costs, not measured whole-grid.
- Accuracy figures in §3.1 are from earlier sessions' recorded runs and were not re-measured today.

Sources: `imageops` / `instances` / `backends` benchmark harnesses, GPU-vs-CPU parity run, and MapLibre/rio-cogeo equivalence checks. All raw figures reproducible from the benchmark scripts.